

# Communication and Runtime Engineering for Two-Party ResNet Inference in HPMPC

ABY2 with HE and silent-OT preprocessing

Technical report, HPMPC `dbg` branch

hpmc 97772ef, ConvTriple fa7b952, flexNN b1ce619

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## Abstract

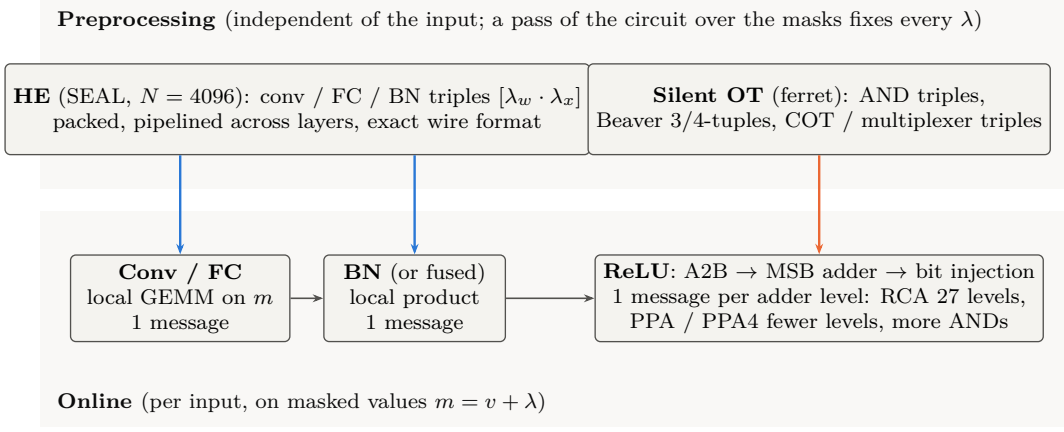
We describe the optimizations and fixes that make HPMPC’s two-party ABY2 inference of ResNet50 fast, reproducible and correct across all of its protocol variants. The work targets both communication and runtime, in the preprocessing phase (homomorphic encryption for linear layers, silent OT for Boolean material) and in the online phase (round latency and per-round work). For one ImageNet image, preprocessing drops from 17.1 s to 4.3 s on two AMD EPYC 7543 hosts (3.3 s on EPYC 9354) and its triple traffic from 1112 to 587 MiB. The online phase drops from 2.4–4.3 s to 0.70 s. BatchNorm triples shrink from 5.7 GB to 0.19 GB of traffic per 10 CIFAR-10 images, and Beaver 3- and 4-tuples need half the OTs and about 45% less traffic. Evaluating several conv-triple chunks at once halves the CIFAR-10 conv phase, and in multi-batch mode the eight lanes of a register share one convolution, which cuts the conv phase per process from 8.6 to 3.9 s and the preprocessing traffic by half. On CIFAR-10 we evaluate 30 protocol variants (three MSB adders, five ReLU families, fused and unfused BatchNorm) in single-batch and 192-image multi-batch mode on two machine pairs. On the EPYC 9354 pair, preprocessing takes 1.3–2.7 s per 10 images and the online phase 0.35–0.50 s; 192 images take 7.8–19.5 s of preprocessing. Public weights, which need no linear-layer triples, and fully shared weight masks (`A_KNOWN=0`) work in every variant. With the AdamW-trained ResNet50, every variant classifies 62–72 of 100 images (plaintext 76) and 118–131 of 192 (61–68%; plaintext 71.9%). Along the way we fixed a thread race, two connection hangs, four protocol bugs that broke whole variant families, and the sources of run-to-run nondeterminism: all 60 threaded builds reproduced their serial reference bit for bit, and the final single-batch builds still give the same outputs.

## 1 Introduction

Two-party inference with a preprocessing phase splits the cost of a secure neural network into two parts. An input-independent *preprocessing phase* produces correlated randomness: multiplication triples for the linear layers, generated with homomorphic encryption (HE), and Boolean triples and tuples for the nonlinear layers, generated with oblivious transfer (OT). The *online phase* then evaluates the network on masked values and needs only a few messages per layer. HPMPC implements this design with an ABY2-style protocol [1] (`PROTOCOL=4`) and generates its preprocessing material with the CHEETAH [2] HE and silent-OT [3] machinery (ConvTriple).

This report collects the engineering work that took this pipeline from a working prototype to a fast and reproducible system. Its focus is communication and runtime; Fig. 2 summarizes the largest effects.

- **HE preprocessing** (Section 3): a packed and pipelined convolution evaluator, an exact wire format that sends only the bits decryption needs, and BatchNorm triples packed into full ciphertexts and generated for all layers in one product, several chunks evaluated at once, and in multi-batch mode one convolution for all lanes. ImageNet preprocessing: 17.1 s  $\rightarrow$  4.3 s (4.0 $\times$ ).
- **Boolean preprocessing** (Section 4): OT packs sized by total demand, ferret parameters and a prefetching LPN step, TCP\_NODELAY on every channel, bit-packed random OTs, and Beaver tuples whose products share one OT per common factor (half the OTs,  $\approx$ 45% less traffic).
- **Online phase** (Section 5): Nagle-free sockets, element-parallel circuit levels for batch size one, pooled GEMM threads, an ahead-of-time PRNG and fewer rounds per ReLU. ImageNet online: 2.4–4.3 s  $\rightarrow$  0.70 s.
- **Correctness and reproducibility** (Section 7): deterministic OT and HE randomness, and fixes for a worker race, lost wake-ups, stale listeners, multi-batch A2B, the threaded GEMM, and the BatchNorm of `A_KNOWN=0` and of public weights.



**Figure 1:** Where the cost of 2PC inference arises. A preprocessing pass fixes every mask; HE produces the linear-layer triples and silent OT the Boolean material. Online, a linear layer costs one message and a ReLU one message per level of its MSB adder.

- **Evaluation** (Section 8): 30 variants  $\times$  2 batch modes  $\times$  2 machine pairs, with final accuracy on a trained model.

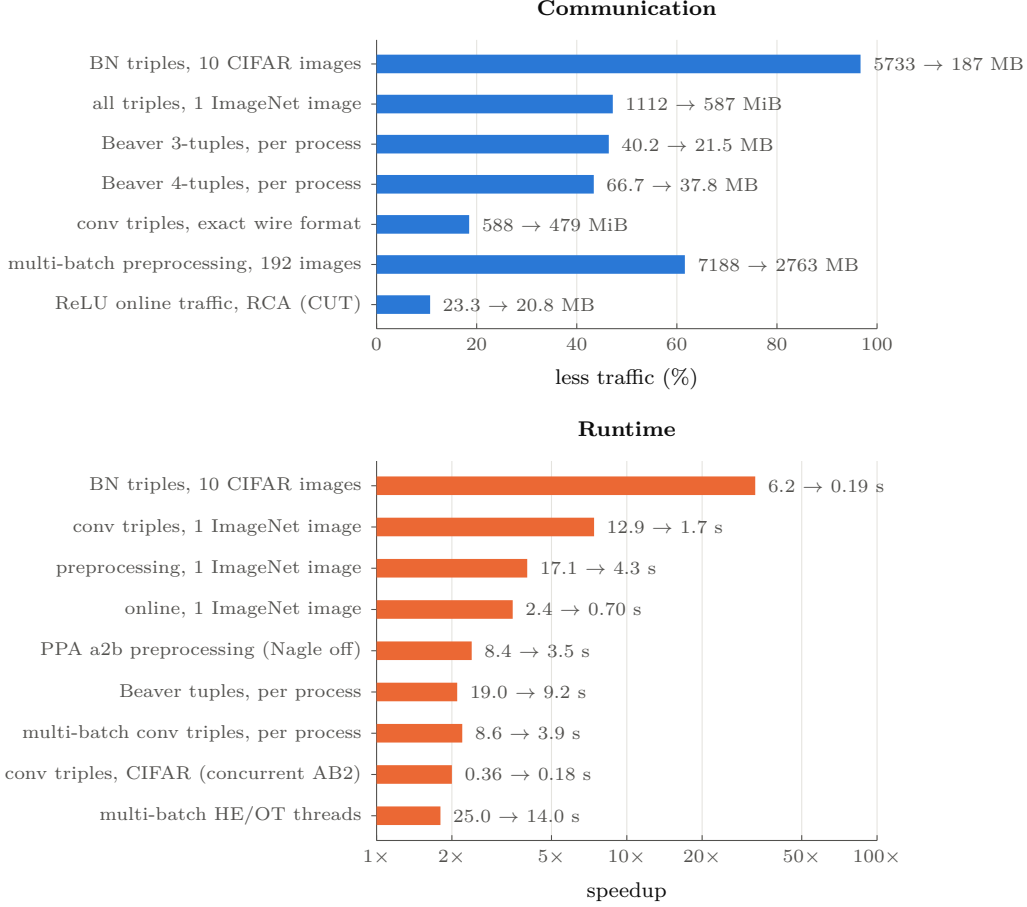
## 2 Setting

**Protocol.** ABY2 shares a value  $v$  as a public masked value  $m = v + \lambda$  together with additive shares of the mask  $\lambda = \lambda_0 + \lambda_1$ . Every mask is fixed in preprocessing, so a linear layer is local except for one message that re-masks its output, and its preprocessing is a triple over the masks, e.g.  $[\lambda_w \cdot \lambda_x]$  for a convolution (Fig. 1). With `A_KNOWN=1` the model owner P0 holds the weight masks and the data owner P1 the activation masks (the AB2 triple); with `A_KNOWN=0` both are shared (AB). A ReLU converts its input to Boolean shares (A2B), extracts the sign with an MSB adder and injects the bit back into the arithmetic domain (bit injection). The adder sets the number of online rounds per ReLU: the ripple-carry adder (RCA) needs 27 AND levels, the parallel-prefix adders PPA and PPA4 (4-input prefix gates, which use Beaver 3- and 4-tuples) need fewer levels but more AND gates.

**Variants.** We measure three adders (RCA, PPA, PPA4) and five ReLU families: *plain*; *reshare* (`RESHARE_OPT`); *reshare+sim*, which folds the reshare material into the convolution mask (`RESHARE_OPT_SIM`); *a2b*, which moves the A2B to preprocessing and bakes its mask into the convolution output (`A2B_ONLINE_OPT`, `A2B_CONV_BAKE`); and *a2b+AKTE*, an adder for inputs whose mask the evaluators know (`A_KNOWN_TO_EVALUATORS_OPT`). Each variant runs with BatchNorm fused into the preceding convolution or unfused. All variants share `BITLENGTH=32`, `FRACTIONAL=5`, probabilistic (SecureML-style) truncation, `FUSE_RELU_AVG=1` and `CUT_FRACTIONAL_BITS_OPT=1`.

**Workloads.** ResNet50 on CIFAR-10 in two modes. *Single batch*: one process, 32-bit lanes (`DATATYPE=32`), 10 images, 32 HE/OT threads and 31 extra online threads. *Multi-batch*: 24 processes with 256-bit registers of eight 32-bit lanes, 192 images, 3 HE/OT threads and one extra online thread per process. The HE preprocessing study and the online steps use ResNet50 on one ImageNet image, `FUNCTION_IDENTIFIER=87`: CHEETAH’s ResNet50 layout for 1000 classes, whose  $224 \times 224$  input arrives padded to  $230 \times 230$ . It runs with a dummy model and image; since no code path depends on the values (Section 3), the times are those of a trained model. Table 1 lists every flag that differs from the defaults in `config.h`.

**Measurement.** Preprocessing and online time are P0’s wall-clock times (the maximum over processes in multi-batch). Single-batch numbers are medians of two runs; multi-batch runs once. All randomness is seeded, so every run of a build produces a bit-identical output; we check each run’s output hash against a serial build without extra threads (Section 7). The runtime does not depend on the weights: HE triples are generated with `TRIPLE_ZERO=ON` and no code path skips zero values. Accuracy uses two trained ResNet50 models: `Cifar_adam_001` for the runtime matrix and an AdamW-trained model (74.35% on the test set) for the final accuracy.



**Figure 2:** Headline effects of the changes in this report, each labeled with its before and after values. Communication: share of traffic removed. Runtime: speedup (log scale). Sections 3–6 explain each.

### 3 Linear layers: HE preprocessing

Figure 3 traces ImageNet preprocessing through the ten ConvTriple changes (s1–s10); Table 3 lists what each step does.

**The two machine pairs.** We ran the s10 build on flare / polynize as well (Table 4). It sends exactly the same bytes, and preprocessing takes 3.31 instead of 4.26 s (22% less). Most of the gap is the convolution (1.13 against 1.74 s): the packed evaluator’s NTTs and multiply-accumulate use AVX-512 on Zen 4 and AVX2 on Zen 3. The ferret key setup and the AND triples, which are bound by memory and AES, gain only 12–14%. The online phase of this build was still dominated by delayed-ACK stalls on both pairs (medians 3.08 and 4.77 s); after the changes of Section 5 it takes 0.69–0.73 s on flare / polynize (1,820 rounds, of which P0 waited 0.15–0.16 s for data). The online steps were measured on flare / polynize only, and the PPA4 adder only there: preprocessing 10.8–11.4 s (before the shared-OT tuples) and online 1.05–1.06 s.

**Packed evaluator (s1, s2).** CHEETAH’s convolution encrypts one input tile per ciphertext and returns one ciphertext per output channel and tile. The packed layout of the GPU path puts several images and output channels into each ciphertext: triple traffic falls from 1112 to 698 MiB. At first this made the evaluator slower (s1), because every weight polynomial still took a full NTT. The evaluator now transforms a weight polynomial right before its only use, and it transforms  $1 \times 1$  weights with NTTs of size  $N/2^a$ . It builds sparse polynomials from a table of powers and sums products lazily in 128 bits. Convolution time falls from 13.7 to 7.5 s.

**Randomness and threads (s3–s6).** Flooding noise and masks come from an AES-CTR generator (AES-NI) instead of SEAL’s Blake2; the thread pool survives across the several parallel calls of a layer; the multiplexer and COT triples run on all threads. Together these steps take preprocessing from 11.9 to 5.4 s.

**Pipelining (s8).** Per layer, each party idled while the other worked: at s7 P0 waited 0.82 s and P1 1.95 s in total. The pipelined protocol (Fig. 4) overlaps P1’s encryption and decryption with P0’s evaluation. It

|                       |                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ImageNet runs         | flags beyond the <code>config.h</code> defaults                                                                                                                                       |
| Model, input          | FUNCTION_IDENTIFIER=87 (default 15), NUM_INPUTS=1, DATATYPE=32 (one image, one process)                                                                                               |
| Preprocessing         | PRE=1, CHEETAH_CONV_PACKED=1 (from s1 on; s0 is CHEETAH’s HomConv2DSS), CHEETAH_CONV_PIPELINE=1 (the default with packing, from s8 on), CHEETAH_THREADS=32                            |
| Network layers        | FUSE_CONV_BN=1, FUSE_RELU_AVG=1, RCA_MSB=1                                                                                                                                            |
| Roles                 | MODELOWNER=P_0, DATAOWNER=P_1 (secret weights: A_KNOWN=1, PUBLIC_WEIGHTS=0, TRUNC_DELAYED=0 and MODELWEIGHTS_KNOWN_DURING_PREPROCESSING=0 were set explicitly, equal to the defaults) |
| ConvTriple (CMake)    | TRIPLE_ZERO=ON; TRIPLE_FERRET=b12 from s10 on (b13 before)                                                                                                                            |
| Online steps (Fig. 8) | in addition CUT_FRACTIONAL_BITS_OPT=1, SEND_BUFFER=100000, RECV_BUFFER=100000, ADDITIONAL_GEMM_THREADS=31, ADDITIONAL_RELU_THREADS=31, RNG_AHEAD=1                                    |

**Table 1:** Build configuration of the ImageNet measurements. The other flags keep their defaults, e.g. `PROTOCOL=4`, `BITLENGTH=32`, `FRACTIONAL=5`.

|               | flare / polynize                                                                         | algofi / goracle                    |
|---------------|------------------------------------------------------------------------------------------|-------------------------------------|
| CPU           | AMD EPYC 9354 (Zen 4), AVX-512, VAES, IFMA                                               | AMD EPYC 7543 (Zen 3), AVX2, AES-NI |
| Cores, memory | 32 cores / 64 threads, 752 GB                                                            | 32 cores / 64 threads, 512 GB       |
| Link          | 25 Gbit/s direct (Intel E810, ice 2.6.7), MTU 9700, 24.7 Gbit/s iperf3, RTT 0.28–0.31 ms |                                     |
| Roles         | P0 (model owner): flare / algofi; P1 (data owner): polynize / goracle                    |                                     |

**Table 2:** Machines. Each host builds its own binaries (`-march=native`).

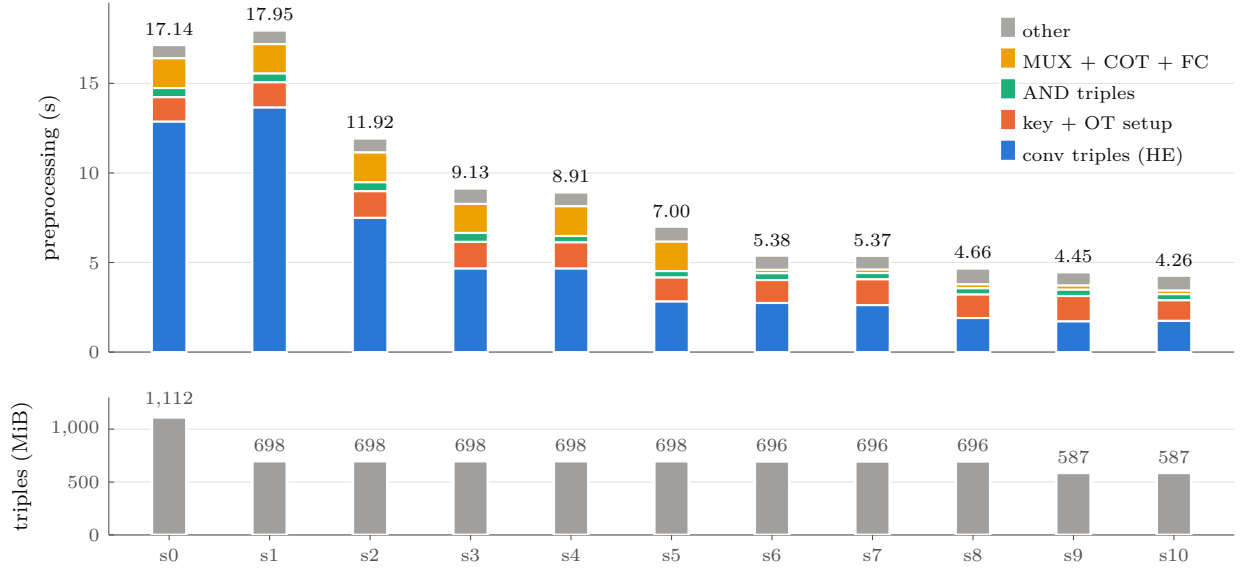
sends exactly the same bytes and cuts convolution time from 2.6 to 1.9 s; with `A_KNOWN=0`, where both parties encrypt, from 3.9–4.0 to 2.0 s in isolation.

**Concurrent evaluation.** Even pipelined, AB2 was slower than AB on CIFAR-10: 0.34–0.39 s against 0.26–0.30 s for the 23 convolutions of the isolated test, although in AB each party evaluates a full convolution under HE. Evaluating one chunk (up to 16 images) is a sequence of parallel regions: loading the inputs, the multiply-accumulate over blocks of output groups and primes, and masking, flooding and truncating the outputs. On CIFAR’s small layers these regions have fewer tasks than threads. An AB party keeps its threads busy by overlapping encryption, evaluation and decryption; the AB2 weight holder evaluated one chunk after another and sent each result before starting the next (Fig. 5). Now three evaluator threads take the chunks in order and a sender thread emits the outputs in chunk order. Chunk  $k$  draws its masks and flooding noise from the PRNG call that the serial loop gave it, so the triples are the same bits and every output hash is unchanged. Table 5 shows the effect of the number of chunks in flight. AB got the same change. End to end, the conv phase of a 10-image preprocessing halves, from 0.54–0.65 to 0.26–0.31 s.

**Exact wire format (s9).** SEAL serializes a seeded input ciphertext with zstd to 62.4 KB. P1 now sends only a 16-byte seed and  $c_0$  at exactly  $60 + 49 = 109$  bits per coefficient (55.8 KB). P0 expands  $c_1$  from the seed directly in NTT form, which saves an inverse and a forward NTT per input. For outputs, the modulus switch before decryption keeps 34 bits of  $c_0$  and 46 of  $c_1$ ; exactly those bits travel, 20–25% less than zstd. The tile search now minimizes bytes rather than ciphertexts, since an input costs about 2.4 times an output. Convolution traffic falls 18% and all preprocessing traffic 12% (Table 6); decrypted values are bit-identical, and a check rejects outputs whose dropped bits are not zero. The 109-bit input modulus is a floor: BFV’s plaintext product carries a term of up to  $\approx 2^{74}$ .

**BatchNorm triples.** Unfused BatchNorm multiplies each activation by a per-channel scale, which the slot-encoded protocol computes as an elementwise product of two vectors. The original code split each image’s elements among the 32 threads before packing: every thread sent its own nearly empty ciphertexts, one image at a time. Packing all images into full ciphertexts cut BN traffic from 2704 / 3029 to 90 / 101 MB and BN time from 6.2–7.9 to 0.7–0.8 s (CIFAR-10, 10 images). Each of the 53 BN layers was still a separate call, and one call is latency-bound (1.6–6.4 MB in 13–50 ms). CHEETAH\_BN\_BATCHED concatenates all slot-encoded layers into one product, split among the threads in whole ciphertexts: the BN phase falls from 0.75 to 0.19 s with the same traffic (Table 7).

**Value independence.** The benchmark’s dummy model makes every weight mask zero, and early versions of the packed evaluator skipped zero weights, so their benchmark times looked 0.6–2.9 s faster. With weights



**Figure 3:** Preprocessing of one ImageNet image on algofi / goracle, by component (median of 3 runs), and the triple traffic sent plus received by P0. s2–s6 are value-independent reruns (Section 3).

| Step | Change                                                                                                                                                                                  | Pre (s) | Conv (s) |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------|
| s0   | Baseline: CHEETAH’s HomConv2DSS, one output ciphertext per channel and tile                                                                                                             | 17.14   | 12.86    |
| s1   | Packed layout: several images and output channels per ciphertext (traffic 1112 → 698 MiB)                                                                                               | 17.95   | 13.65    |
| s2   | Packed evaluator: weight polynomials transformed right before use, $1 \times 1$ weights through NTTs of size $N/2^a$ , sparse polynomials from a power table, lazy 128-bit accumulation | 11.92   | 7.49     |
| s3   | AES-CTR noise and masks instead of Blake2; ciphertexts loaded in parallel                                                                                                               | 9.13    | 4.66     |
| s4   | AND triples from bit-packed random OTs (64 per word)                                                                                                                                    | 8.91    | 4.66     |
| s5   | Thread pool kept across calls; multiply-accumulate blocked per prime                                                                                                                    | 7.00    | 2.82     |
| s6   | Multiplexer and COT triples on all threads (MUX 1.08 → 0.07 s, COT 0.48 → 0.04 s)                                                                                                       | 5.38    | 2.74     |
| s7   | All work independent of the weights’ values (TRIPLE_ZERO=ON)                                                                                                                            | 5.37    | 2.62     |
| s8   | Convolutions pipelined across layers (Fig. 4)                                                                                                                                           | 4.66    | 1.90     |
| s9   | Exact wire format and byte-optimal tiling (conv traffic 588 → 479 MiB)                                                                                                                  | 4.45    | 1.71     |
| s10  | Prefetching LPN step, ferret parameters b12 (key setup 1.42 → 1.13 s)                                                                                                                   | 4.26    | 1.74     |

**Table 3:** The HE and OT preprocessing steps (ImageNet, algofi / goracle, 32 threads). Table 4 compares the s10 build on both machine pairs.

known in preprocessing, a zero-skipping evaluator would also leak the number of zero weights through its timing. All evaluators now do the same work for any weight values, and all numbers in this report are value-independent.

## 4 Boolean material: OT preprocessing

The ReLUs consume AND triples, Beaver 3- and 4-tuples (PPA4), COT triples (bit injection) and multiplexer triples. All come from ferret silent OT, 32 OT packs of two ferret instances each.

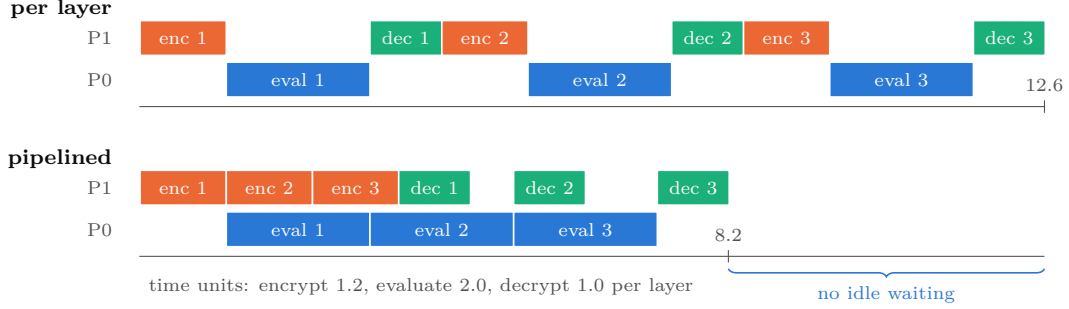
**Ferret parameters and the LPN step.** With 32 instances at once, ferret’s LPN step is memory-bound on its 7.2 MB tables. Computing the AES of the next four groups ahead and prefetching their table rows cuts the step from 0.34 to 0.28 s with bit-identical outputs. Parameter set b12 has a 3.8 MB table that stays in cache, for 6.5 MB more setup traffic per direction. Key setup falls from 1.42 to 1.13 s. VAES did not help: the LPN step is memory-bound and MITCCRH is bound by its AES key schedule.

**Demand-sized OT packs.** The first OT consumer used to size the packs, and later consumers extended them in small steps. hpmc now passes the total OT demand of the whole preprocessing to ConvTriple (ot\_demand\_hint) before the first generation. On CIFAR-10 the OT-pack phase falls from 0.91 to 0.47 s, and PPA4, reshare and a2b get enough capacity for their tuples.

**TCP\_NODELAY everywhere.** The ConvTriple channels enabled TCP\_NODELAY only in WAN mode. The COT multiply behind the a2b bake flushes small messages 31 times in sequence, and each flush waited for a delayed

| ImageNet, one image, s10 build (median of 3 runs) | algori / goracle (Zen 3) | flare / polynize (Zen 4) | ratio |
|---------------------------------------------------|--------------------------|--------------------------|-------|
| Preprocessing (s)                                 | 4.26                     | 3.31                     | 1.29  |
| key setup (ferret, HE keys)                       | 1.15                     | 1.01                     | 1.13  |
| conv triples                                      | 1.74                     | 1.13                     | 1.55  |
| AND triples                                       | 0.34                     | 0.29                     | 1.17  |
| multiplexer / COT / FC triples                    | 0.07 / 0.03 / 0.10       | 0.08 / 0.07 / 0.08       |       |
| Triple traffic, P0 sent / received (MB)           | 305.5 / 281.9            | 305.5 / 281.9            | 1     |
| Online (s), before the online changes             | 4.77                     | 3.08                     |       |
| Online (s), final online build (Fig. 8)           | –                        | 0.69–0.73                |       |

**Table 4:** The same ImageNet build on both machine pairs. The online times of the s10 build vary with delayed-ACK stalls (flare / polynize 2.43–3.21 s over three runs).



**Figure 4:** Pipelining the convolutions across layers (s8). Per layer, P1 waits while P0 evaluates and vice versa; pipelined, P1 encrypts the next layer’s inputs and decrypts the previous layer’s outputs while P0 evaluates. Same bytes, preprocessing 5.69 → 4.66 s.

ACK. One call took 170 ms; with Nagle off it takes 11 ms. PPA a2b single-batch preprocessing falls from 8.4 to 3.5 s, PPA4 reshare+sim from 8.9 to 5.0 s. The same fix on hpmc’s own sockets takes the ImageNet online phase from 2.4–4.3 to 1.9–2.2 s (Section 5).

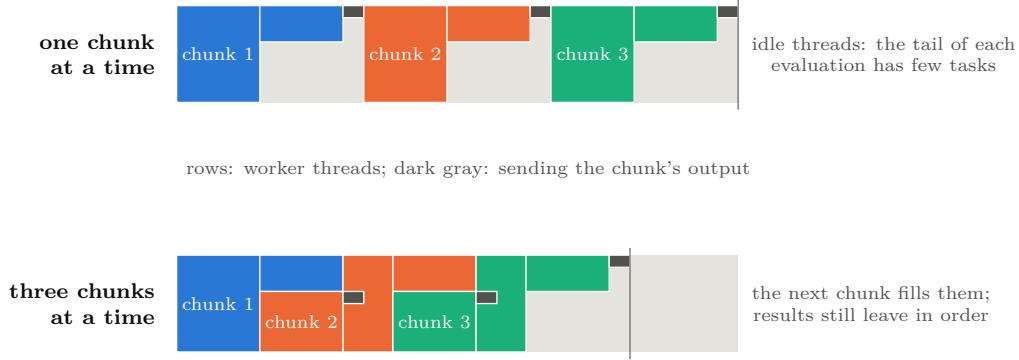
**Bit-packed random OTs.** Random OTs with one-bit messages are hashed and reduced to bits in cache-sized chunks and stored eight per byte (`send/recv_rot_bits`), where the generic interface kept two 16-byte blocks per OT. AND triples take 0.36 instead of 0.50 s, and the COT multiply reads the same bits with far less memory traffic.

**Beaver 3- and 4-tuples with shared OTs.** A 3-tuple  $(a, b, c, ab, ac, bc, abc)$  was generated as an AND triple  $(a, b, ab)$  plus three independent share multiplications  $a \cdot c$ ,  $b \cdot c$  and  $ab \cdot c$ . Each multiplication took one random OT per direction: the receiver sent a choice correction  $e = s \oplus x$ , the sender a masked value  $y \oplus r_0 \oplus r_1$ . The three products share the factor  $c$ . A random OT’s messages are hashes of 128-bit blocks, so one OT can carry  $k$  message bits for the price of one (`send/recv_rot_bitplanes`). `cot_outer_multiply` computes all  $x_p \cdot y$  for  $k$  values  $x_p$  from one OT per direction, sending  $k$  masked bits and at most one correction bit. Because  $c$  is fresh, each party takes its share of  $c$  from the random choice bits of its receiving OTs and sends no correction at all (Fig. 6). A 4-tuple needs the nine products  $\{a, b, ab\} \times \{c, d, cd\}$ : three OTs per direction instead of nine. Per tuple, AND triples included, a 3-tuple now needs 4 random OTs instead of 8 and a 4-tuple 10 instead of 22. In a multi-batch PPA4 run the tuple phases fall from 19.0 to 9.2 s per process and their traffic by 43–46% (Fig. 7). The party-local variant used by reshare+sim, where P0 holds  $b$  and P1 holds  $c$  in full, is covered by the same code with one correction bit for P0’s zero share. Unit tests check all products; ReLU exactness tests report 0 wrong outputs.

## 5 Online phase

On a 0.3 ms LAN, the online phase is bound by round latency and per-round work, not by bandwidth. Activations send about 180 MB per ImageNet inference, 0.06 s at 25 Gbit/s. Figure 8 shows the steps for one ImageNet image.

**Nagle, buffers and GEMM threads.** With Nagle’s algorithm, a round’s last small message waited for a delayed ACK. ReLU layers were bimodal (6 or 46 ms), and the online phase varied between 2.4 and 4.3 s. With



**Figure 5:** Why concurrent evaluation helps (schematic). The tail of each evaluation has fewer tasks than threads; with several chunks in flight, the next chunk fills the idle threads.

| isolated pipelined suites, flare / polynize, 32 threads<br>chunks evaluated at once | AB2 (A_KNOWN=1) |              | AB (A_KNOWN=0) |              |
|-------------------------------------------------------------------------------------|-----------------|--------------|----------------|--------------|
|                                                                                     | CIFAR (s)       | ImageNet (s) | CIFAR (s)      | ImageNet (s) |
| 1, sending in the evaluation loop (before)                                          | 0.34–0.39       | –            | 0.26–0.30      | 1.03         |
| 1, separate sender thread                                                           | 0.27–0.29       | 0.94         |                |              |
| 2                                                                                   | 0.19–0.20       | 0.77         |                |              |
| 3 (default)                                                                         | 0.18–0.19       | 0.73         | 0.21           | 0.82         |
| 4                                                                                   | 0.17–0.18       | 0.75         |                |              |

**Table 5:** Conv triples with several chunks in flight (CONV\_PIPE\_EVALUATORS); every output checked exactly. Traffic is unchanged: CIFAR 99 MiB (AB2) and 197 MiB (AB), ImageNet 479 and 958 MiB.

TCP\_NODELAY it takes 1.9–2.2 s. Send and receive buffers of 100 000 words mean fewer messages and thread hand-offs per round. The threaded GEMM first sent its outputs in linear order while the completion visits  $64 \times 64$  tiles; CIFAR-10 accuracy was 0% until both used the tile order. Its threads now live in a lock-free pool.

**Element-parallel levels at batch size one.** HPMPc vectorizes only across independent inputs, so a single image evaluated every level of a ReLU (adder steps, A2B, bit injection) element by element on one thread. Within a level, every element runs the same code and consumes the same number of words from each stream: sends, receives, preprocessed values, its own random words and the indices into the triple buffers. `stream_parallel_for` measures this consumption on the first element and runs the rest on the worker pool, each worker with cursors at exactly the stream positions the serial loop would use (Fig. 9). Segments end at send/receive buffer boundaries, and a misuse check verifies every worker’s consumption. The messages, their order and all values are those of the serial loop. With bulk random words, parallel adder construction and the GEMM mask-and-send and completion on the same pool, the ImageNet ReLU time falls from 630 to 300 ms and the online phase from 1.09 to 0.70 s. `RNG_AHEAD` moves the party’s own sequential AES stream to a background thread.

**Rounds per ReLU.** What remains is round latency: 1,752 rounds for CIFAR-10 with RCA against 776 with PPA and 625 with PPA4 (Table 8). `CUT_FRACTIONAL_BITS_OPT` removes the fractional bits from the MSB adder by substituting identity elements and public constants. RCA loses 245 of its 1,997 rounds (12%), and ReLU traffic falls 8–11% for every adder. Preprocessing and accuracy do not change measurably (1,134 vs. 1,138 of 1,752 evaluations correct with and without CUT).

## 6 Multi-batch

With 24 processes of 256-bit lanes, preprocessing dominates. With more than one HE/OT thread per process, the processes shared connections: channel  $i$  used port  $base + i \cdot PROCESS\_NUM$  while process bases stepped by 3, which caused hangs and wrong triples. With a stride of  $3 \cdot PROCESS\_NUM$ , preprocessing falls from 25.0 to 14.0 s with three threads per process (Table 10). One extra online thread per process cuts the online phase from 0.81 to 0.67 s with an identical output hash.

**Lanes in one convolution.** Each 256-bit register holds eight images (lanes), and the conv triples were generated as one single-image convolution per lane and layer, each with its own copy of the weight masks: 424

| MiB, P0 sent / received | before (s8)           | after (s10)           |
|-------------------------|-----------------------|-----------------------|
| Conv triples            | 232.6 / 355.1         | 234.9 / 244.1         |
| Ferret setup            | 46.2 / 46.2           | 52.7 / 52.7           |
| Multiplexer triples     | 35.4 / 35.4           | 35.4 / 35.4           |
| COT triples             | 34.4 / 1.1            | 34.4 / 1.1            |
| FC triples              | 0.7 / 1.2             | 0.7 / 1.2             |
| Total                   | 349.4 / 439.0 (788.4) | 358.2 / 334.6 (692.8) |

**Table 6:** Traffic of one ImageNet preprocessing ( $A\_KNOWN=1$ ). With  $A\_KNOWN=0$  the triple traffic falls from 659 / 626 to 551 / 518 MiB.

| CIFAR-10, 10 images          | traffic P0 sent / recv (MB) | BN phase (s) | preprocessing (s) |
|------------------------------|-----------------------------|--------------|-------------------|
| threads split each image     | 2704 / 3029                 | 6.2–7.9      | 8.2–10.0          |
| full ciphertexts             | 90 / 101                    | 0.7–0.8      | 3.0–3.2           |
| all 53 layers in one product | 88 / 99                     | 0.19         | 1.94 (RCA)        |

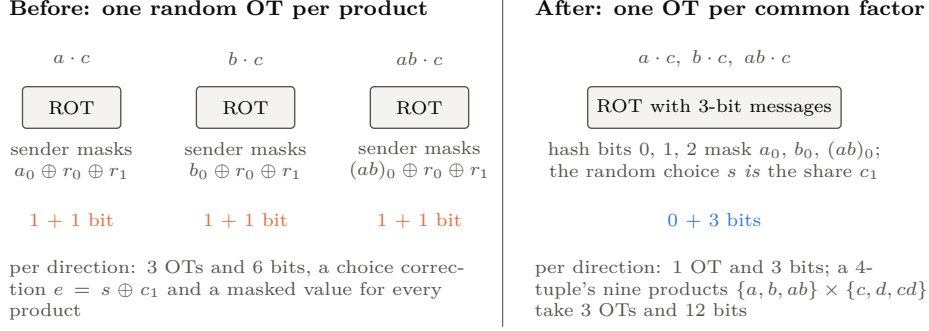
**Table 7:** BatchNorm triples with unfused BN. In multi-batch the per-process BN phase falls from 1.16 to 0.85 s (3 threads per process, compute-bound).

small, unpipelined convolutions per process, with ciphertexts that are mostly empty. The model owner’s mask of a weight is  $-w$  (SHARE\_PREP), and every lane’s image uses the same weights, so all lanes have the same weight masks by construction. The eight triples of a layer are therefore one convolution over eight images with one set of weights. CHEETAH\_CONV\_LANES generates them that way and passes all layers to one pipelined call. A debug check confirmed that no lane’s masks differ in any layer or process. With three HE/OT threads per process the conv phase falls from 8.4–8.7 to 3.8–3.9s and P0’s preprocessing traffic by 50–62% (Table 9); four threads, which enable the pipeline, gain nothing more. With the AdamW model the accuracy is unchanged (PPA 121 of 192 against 123 without, RCA 131, PPA4 127). The premise does not depend on  $A\_KNOWN$ : input sharing gives the model owner the share  $-w$  and the other party 0 in every lane either way, and  $A\_KNOWN=0$  only turns the triple into an AB triple, for which both parties encrypt their shares. The lane batching therefore also applies to  $A\_KNOWN=0$ . In a local test (one process, 8 lanes of 4 images, 4 HE threads, AdamW model) the lane check stays silent for both parties, the conv phase falls from 12.9 to 7.2s and the conv traffic from 1036 to 540 MB per direction, and 22 against 23 of 32 images are classified correctly.

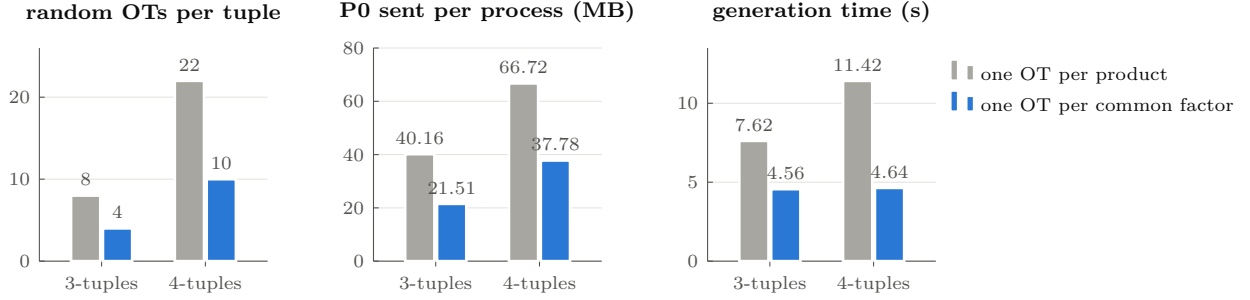
## 7 Correctness and reproducibility

Fast code is only useful if its outputs are right and repeatable. We made every run bit-reproducible, compared each threaded build against a serial one and bisected every configuration that lost accuracy over the added optimizations; this exposed the following bugs.

- **Nondeterministic OT.** COT and multiplexer triple shares are the ferret outputs, and hpmc truncates shares locally, so their per-run randomness changed the outputs. ConvTriple’s PRGs are now seeded per party, stage, call and task, and a scoped override seeds emp’s default PRGs per OT pack and worker (DetSeedScope).
- **Worker race in P0’s A2B.** For the reshare+sim skip with PPA4, P0 chooses each group’s slice masks from the Beaver 3-tuples that the group’s adder will consume, found through a count of groups prepared so far. On the worker pool that count was one shared global. Threaded PPA4 reshare+sim with fused BN gave a different output every run. The count is now a per-worker stream index. The race stayed hidden whenever the pool threads woke one after another.
- **Hangs.** A lost wake-up at connection start, where flag and broadcast were not under the waiters’ mutex, and a stale listener into whose backlog the next phase’s peer connected.
- **Multi-batch a2b.** The A2B mask bake derived the conv mask by casting transposed words to lanes, which holds only for 32-bit lanes; with 256-bit registers the a2b families classified 13–29 of 192 images. The mask is now computed value by value with the A2B’s own inverse packing, which leaves single-batch hashes unchanged, and the multi-batch a2b families classify 99–132 of 192 images, like the plain family.
- **Threaded GEMM with  $A\_KNOWN=0$  or public weights.** The threaded GEMM had diverged from the serial one. With  $A\_KNOWN=0$  it sent the outputs without the conv triple’s share and left the triple cursor behind, so every threaded build classified at chance (1 of 10). With public weights and TRUNC\_DELAYED=0



**Figure 6:** Products with a common factor share one random OT. Before, each product  $x \cdot c$  took one random OT per direction plus two sent bits. Now one OT per direction carries all  $k$  products in its  $k$  message bits, and a party whose share of  $c$  is fresh takes it from the OT’s random choice bit.



**Figure 7:** Beaver tuple generation per process in a multi-batch PPA4 run (flare / polynize, 24 processes).

it truncated the first layer with the regular truncation while the completion expected the data owner’s exact one, and both parties waited forever in preprocessing. It now follows the serial path.

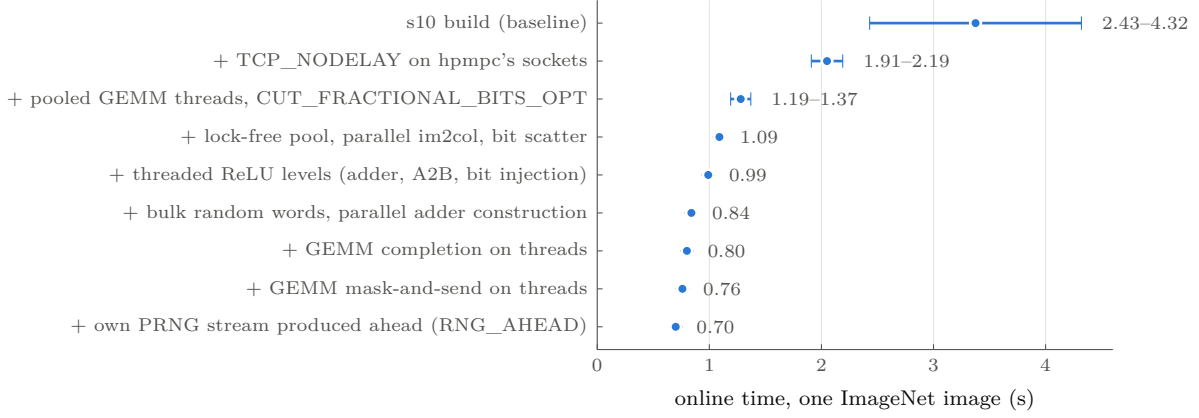
- **BatchNorm triples with A\_KNOWN=0.** AB triples multiply the full masks, and the BN product’s operand is  $x - \mu$ , whose mask includes the model owner’s mask of  $\mu$ . The triple was set up over the masks of  $x$ , which left out  $\lambda_s \lambda_\mu = s\mu$ : every output was  $s \cdot x + \beta$  (unit test failed, ResNet50 unfused 0 of 10). With A\_KNOWN=1 only the data owner’s mask share enters, and  $\mu$  does not change it.
- **Public weights with unfused BN and delayed truncation.** With public weights the first convolution’s output is still the data owner’s sharing (the other party’s mask is zero). With TRUNC\_DELAYED=1, with or without BIT\_INJECTION\_TRUNC\_SIM, the BN truncated it with the SecureML local truncation, which wraps for positive values (0–1 of 10 correct). The owner now truncates it exactly and re-masks, in the same single message.
- **Test harness.** The ReLU tests fed unbaked inputs to the reshare+sim skip (35% wrong); the harness now declares its inputs unbaked.

After these fixes, all 60 threaded variants (30 single-batch, 30 multi-batch) reproduced their serial build’s output hash on flare / polynize (round 4), and repeated runs are bit-identical. The concurrent conv-triple evaluation keeps every single-batch hash of round 4. The lane batching gives the multi-batch triples different random shares, so those hashes changed; its correctness rests on the lane check and on the accuracy (Section 6 and Fig. 13). Hashes differ between the machine pairs, because the PRNG draws 512-bit AES blocks with VAES-512 on Zen 4 and 256-bit blocks on Zen 3.

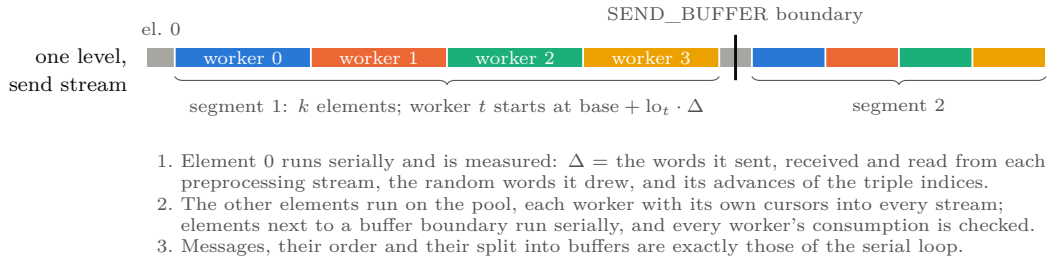
## 8 Evaluation of all variants

Figure 10 shows preprocessing and online time for every variant on both machine pairs.

**Single batch.** On flare / polynize, preprocessing takes 1.3–1.8 s with RCA, 1.9–2.2 s with PPA and 2.1–2.5 s with PPA4 when BN is fused, and 1.6–2.0, 2.2–2.4 and 2.3–2.7 s unfused; the a2b families sit at the upper end. The online phase takes 0.35–0.50 s. Because it is round-latency-bound, PPA and PPA4 are slightly faster online than RCA despite more AND gates. The fastest configuration in total is RCA with fused BN (1.32 s + 0.39 s).



**Figure 8:** Online phase of one ImageNet image on flare / polynize (ranges over runs where they varied). Output hashes are unchanged by every step.



**Figure 9:** `stream_parallel_for`: element-parallel circuit levels at batch size one. One element is measured; the level's remaining elements run on the worker pool with per-worker cursors into every stream, split at buffer boundaries.

**Multi-batch.** 192 images take 7.8–11.7 s (RCA), 11.0–15.0 s (PPA) and 13.9–19.5 s (PPA4) of preprocessing and 0.54–0.81 s online. Per image that is 41–101 ms of preprocessing, against 132–267 ms in single batch. The a2b+AKTE family is the fastest PPA4 variant here (13.9–14.8 s): its adder needs fewer tuples.

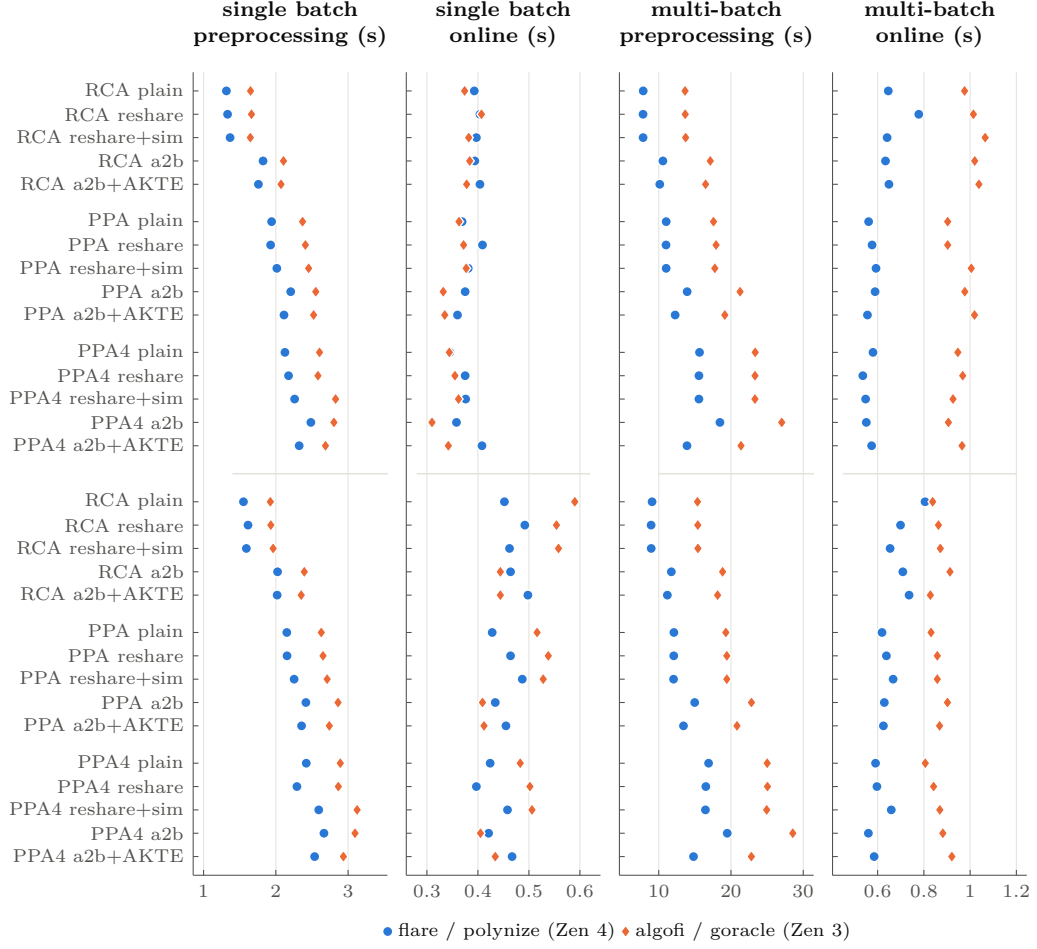
**Machines.** In single batch, algo / goracle needs  $1.13\text{--}1.25\times$  the preprocessing time of flare / polynize and  $0.84\text{--}1.30\times$  the online time. Multi-batch is compute-bound, and there Zen 3 needs  $1.46\text{--}1.75\times$  as long for preprocessing and  $1.04\text{--}1.84\times$  online: it lacks AVX-512 for the HE NTTs and the GEMM. The lane batching gains less there (conv phase per process  $14.8 \rightarrow 8.0$  s against  $8.4 \rightarrow 3.8$  s; preprocessing 20–35% less against 19–38%). ImageNet shows the same pattern (Table 4).

**The last optimizations.** Figure 11 separates the steps. The shared-OT tuples and the batched BN triples (round 2 to round 4) help PPA4 most: single batch  $4.35 \rightarrow 2.68$  s unfused and  $3.46 \rightarrow 2.61$  s fused, multi-batch  $31.5 \rightarrow 22.6$  s and  $30.2 \rightarrow 20.2$  s; unfused RCA gains  $2.51 \rightarrow 1.94$  s from the batched BN. The concurrent conv-triple evaluation then saves 0.26–0.55 s in every single-batch variant (10–26%), and the lane batching 4.5–6.3 s in every multi-batch variant (19–38%). From round 2 to the final code, preprocessing of the plain family falls by 17–44% in single batch and by 30–48% in multi-batch.

## 8.1 Fully shared masks and public weights

Two further settings change who holds the weights. With `A_KNOWN=0` both parties hold shares of the weight masks as well as of the activation masks, so the conv, FC and BN triples are AB triples for which both parties encrypt. With public weights (`PUBLIC_WEIGHTS=1`) the linear layers multiply locally and need no triples at all. We run public weights with `TRUNC_DELAYED=1` and `BIT_INJECTION_TRUNC_SIM=1`: each linear layer's truncation is deferred and folded into the next ReLU's bit injection instead of costing a round of its own. Figure 12 compares both settings with the secret-weight default for the plain family.

**Public weights.** Only the ReLU material remains to be preprocessed: across all 30 variants, preprocessing takes 0.92–2.03 s per 10 images and 3.5–14.3 s per 192 images, 16–41% and 22–61% less than the same variant with secret weights in the final builds. The online phase takes 0.27–0.41 s and 0.33–0.42 s (9–33% and 30–50% less): convolutions send nothing, and the truncation shares the bit injection's message. With the AdamW



**Figure 10:** All 30 variants, CIFAR-10 ResNet50 (final builds). Upper block: fused BN; lower block: unfused BN. Single batch: 10 images, median of two runs. Multi-batch: 192 images, 24 processes. Tables with every value: `docs/VARIANT_PERFORMANCE.md`.

| single batch, unfused BN | rounds |        | ReLU traffic (MB) |        | online (s) |        |
|--------------------------|--------|--------|-------------------|--------|------------|--------|
|                          | CUT    | no CUT | CUT               | no CUT | CUT        | no CUT |
| RCA                      | 1,752  | 1,997  | 20.8              | 23.3   | 0.52       | 0.48   |
| PPA                      | 776    | 789    | 32.6              | 36.8   | 0.44       | 0.49   |
| PPA4                     | 625    | 625    | 20.1              | 21.8   | 0.42       | 0.42   |

**Table 8:** Online rounds, P0’s ReLU traffic and online time for 10 CIFAR-10 images on flare / polynize, with and without CUT\_FRACTIONAL\_BITS\_OPT. The time differences are within run-to-run noise.

| multi-batch, 192 images, flare / polynize | preprocessing (s) |       | conv phase per process (s) |       | P0 sent (MB)<br>before → lanes |
|-------------------------------------------|-------------------|-------|----------------------------|-------|--------------------------------|
|                                           | before            | lanes | before                     | lanes |                                |
| RCA, unfused                              | 13.7              | 8.9   | 8.64                       | 3.87  | 8,884 → 4,459                  |
| RCA, fused                                | 12.5              | 8.0   | 8.43                       | 3.86  | 7,188 → 2,763                  |
| PPA, unfused                              | 16.9              | 12.0  | 8.73                       | 3.88  | 9,022 → 4,597                  |
| PPA4, unfused                             | 22.6              | 16.7  | 8.67                       | 3.82  | 10,238 → 5,813                 |

**Table 9:** Conv triples of all lanes as one convolution (CHEETAH\_CONV\_LANES), plain family, 3 HE/OT threads per process.

model, the public-weight variants classify 63–73 of 100 images in single batch and 115–131 of 192 (60–68%) in multi-batch. Unfused BatchNorm needed the first-layer truncation fix of Section 7; before it, those variants classified 0 of 10 images.

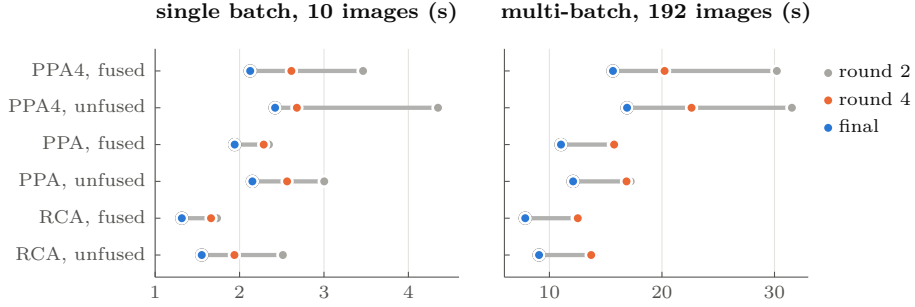
**Fully shared masks.** With `A_KNOWN=0` the linear-layer triples are AB triples, and the preprocessing traffic grows by a factor of 1.4–1.9 (RCA, unfused BN, 10 images: 229 to 428 MB). In single batch, preprocessing takes 1.34–2.74 s over the 30 variants and the online phase 0.34–0.52 s. Since the concurrent evaluation (Section 3), `A_KNOWN=1` is faster in both HE phases of every variant (conv 0.27–0.33 against 0.30–0.39 s, BN 0.20–0.24 against 0.23–0.26 s); the totals differ by  $-0.07$  to  $+0.13$  s, within the noise of the OT phases. In multi-batch, measured before the lane batching was enabled for it (Section 6), `A_KNOWN=0` takes 13.8–26.4 s, 5.8–7.1 s (1.3–1.8 $\times$ ) more than `A_KNOWN=1` with lanes. With the AdamW model `A_KNOWN=0` classifies 64–72 of 100 images in single batch and 117–133 of 192 in multi-batch. Its accuracy first looked like chance level: a bisection over the added optimizations found the threaded GEMM (with the extra threads off, 5 of 10 correct) and, for unfused BN, the BatchNorm triples, both described in Section 7.

## 8.2 Final accuracy

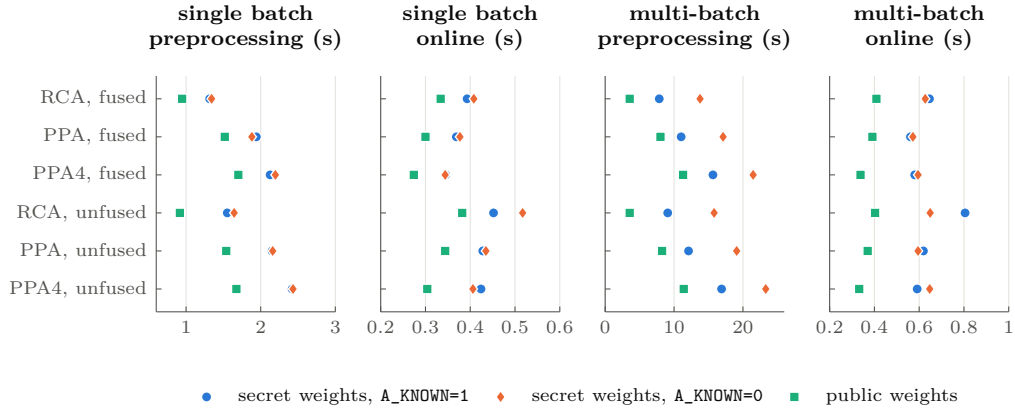
Every variant stays above 60% (Fig. 13 and Table 11). The spread between variants (62–72 of 100, 118–131 of 192) is truncation noise: the variants draw different amounts of randomness, so different values wrap. The adders are exact, with 0 of 32,768 random ReLU inputs wrong for each adder. Fused and unfused BN are equally accurate with this model. With the older `Cifar_adam_001` model, probabilistic truncation in the 32-bit ring cost 10–17 points (Trio 56% against 73% in plaintext; 73% with a 64-bit ring). The AdamW model is much less sensitive to it.

| HE/OT threads | preprocessing (s) | conv (s) | AND triples (s) | BN (s) | online (s) |
|---------------|-------------------|----------|-----------------|--------|------------|
| 1             | 25.0              | 16.5     | 3.6             | 2.1    | 0.63       |
| 2             | 17.0              | 10.2     | 3.3             | 1.4    | 0.68       |
| 3             | 14.0              | 8.8      | 3.0             | 1.2    | 0.75       |
| 4             | 14.0              | 7.8      | 3.8             | 1.0    | 0.69       |

**Table 10:** Multi-batch CIFAR-10 (24 processes, 192 images, RCA, unfused BN) on flare / polynize, by HE/OT threads per process, after the port-stride fix.



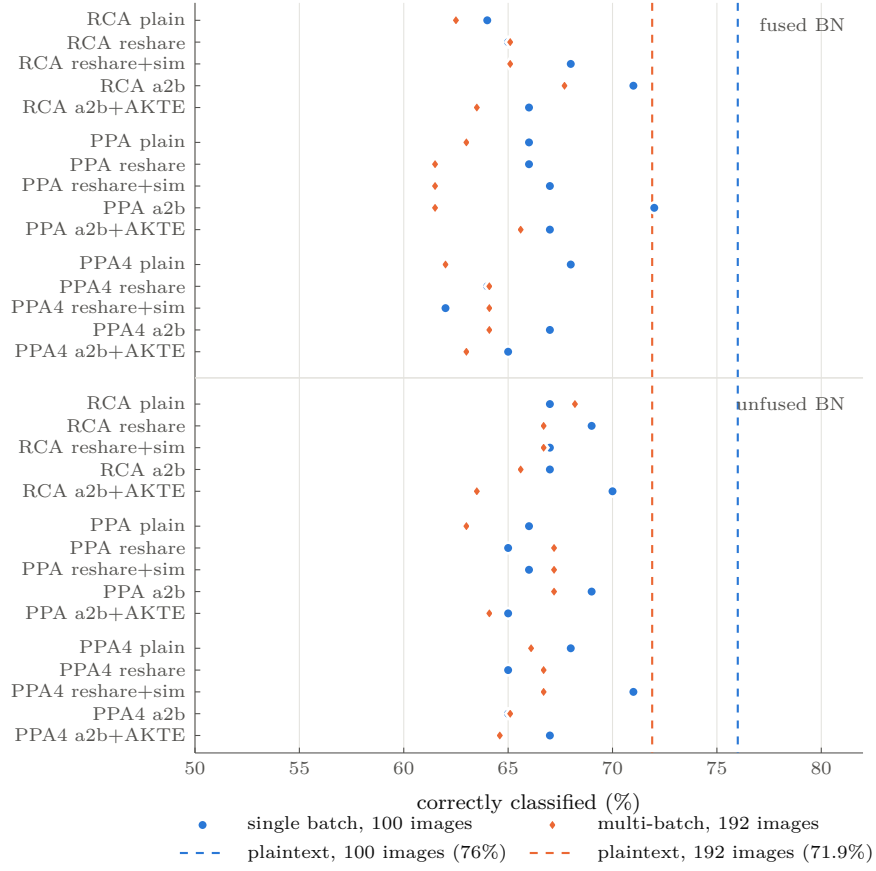
**Figure 11:** Preprocessing of the plain family in round 2, round 4 (after the shared-OT tuples and the batched BN triples) and with the final code (concurrent conv-triple evaluation; multi-batch: lane batching), flare / polynize.



**Figure 12:** Secret weights with  $A\_KNOWN=1$  (default) and  $A\_KNOWN=0$ , and public weights, plain family, flare / polynize (final builds; multi-batch  $A\_KNOWN=0$  before the lane batching was enabled for it, public weights have no conv triples).

| Setting                                                    | images    | correct          |
|------------------------------------------------------------|-----------|------------------|
| Plaintext, float (PyTorch)                                 | 100 / 192 | 76 / 138 (71.9%) |
| Trio 3PC, standard configuration                           | 100       | 71               |
| Trio 3PC, reduced-slack truncation ( $TRUNC\_APPROACH=1$ ) | 100       | 70               |
| 2PC ABY2, single batch, all 30 variants                    | 100       | 62–72            |
| 2PC ABY2, multi-batch, all 30 variants                     | 192       | 118–131          |
| 2PC ABY2, $A\_KNOWN=0$ , single / multi-batch              | 100 / 192 | 64–72 / 117–133  |
| 2PC ABY2, public weights, single / multi-batch             | 100 / 192 | 63–73 / 115–131  |
| 2PC ABY2, CUT on / off (plain family, 12 runs each)        | 1,752     | 1,134 / 1,138    |

**Table 11:** Final accuracy with the AdamW-trained ResNet50 on the first CIFAR-10 test images.



**Figure 13:** Accuracy with the AdamW-trained ResNet50, standard probabilistic truncation, final builds on flare / polynize: 100 images in single batch, 192 in multi-batch (lane-batched conv triples).

## 9 Limitations and open problems

- **Multi-batch A\_KNOWN=0 on the testbed.** The lane batching applies to A\_KNOWN=0 as well (Section 6), but the multi-batch A\_KNOWN=0 numbers of Fig. 12 were measured before it was enabled there; only the local test above covers it.
- **Truncation.** ABY2 has no reduced-slack or exact truncation yet. It is not needed for accuracy with the AdamW model, but it would remove the truncation noise (Section 8).
- **Weights known in preprocessing.** The multi-batch a2b bake still excludes MODELWEIGHTS\_KNOWN\_DURING\_PREPROCESSING, whose prescribed shares assume one word per register.
- **Ciphertext occupancy.** At  $7 \times 7$  feature maps, inputs are 5–19% data and outputs 6–24%. Packing the sparse outputs needs automorphisms and a larger ring ( $N = 8192$ ).
- **OT extension.** Ferret’s LPN step and the AES key schedules of the correlation-robust hash are now the largest CPU costs of the Boolean preprocessing; a fixed-key hash is a security decision.

## 10 Conclusion

The largest runtime gains came from doing less redundant work in the HE evaluator (packing, lazy transforms, pipelining). The largest communication gains came from sending only information-bearing bits: the exact wire format, full BatchNorm ciphertexts, and OTs whose message bits carry several products. In the online phase, latency rather than bandwidth dominates, so the gains came from Nagle-free sockets, fewer rounds and parallelizing the per-round work at batch size one. Seeded, bit-reproducible runs made every change measurable and exposed the bugs in Section 7.

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